Change/Configuration Management

At the end of this episode, I will be able to:

1. Explain typical change management processes used in organizations today
2. Explain typical configuration management processes used today

Learner Objective: *Learn about change management and configuration management in networking today*

Description: In this episode, you will learn about the important topics of change management and configuration management used in organizations today. You will learn to define these areas as well as discuss typical components and processes used in their function.

- Configuration management - configuration management refers to the process of efficiently handling and controlling changes to a network's configuration settings, devices, and software. The goal is to maintain a stable and reliable network environment while allowing for necessary updates, modifications, and expansions. This concept is crucial for ensuring consistency, security, and scalability in large and complex network infrastructures. Key aspects of this discipline include:

- Device configurations - it is imperative that a system be in place to manage the many device configurations required in a modern network; remember, device configurations would include things like servers, routers, switches, firewalls, and many many more. This system for storing them should have great security features and a great system for check-in and check-out, etc.

    - Production configuration - one type of configuration that needs to be managed is the production configurations. These are the configurations that are actually in use in the organization.
    - Backup configuration - taking configuration backups is often a requirement in configuration management environments, therefore, you must accommodate the management of these backup configurations. You might also be responsible for multiple generations of backup configurations.
    - Baseline/golden configuration - other common types of configurations you might be responsible for include baseline and golden configurations. A baseline configuration is the standardized and well-defined set of initial settings, parameters, or states for a system or device, serving as a reference point from which changes and deviations can be measured and managed. A golden configuration is the meticulously optimized and security-hardened baseline setting for a system or device, representing the ideal and secure state that serves as the standard for deployment across a network or infrastructure.

- Change Management - is another important aspect of a configuration management system. This aspect deals with implementing a systematic approach to handling changes in the network, including additions, modifications, and removals. Your obsession here should be with documenting and tracking changes to understand the impact on the network and to assist in troubleshooting issues. This process will often involve a well-documented service request and request process tracking procedure. There are plenty of enterprise-level software packages designed to assist with this area.
- Other aspects of configuration management include:

- Version control - as mentioned above, your system might include the maintenance of many configuration versions, permitting the ability to rollback to various states of the device or software.
- Automation - many configuration management systems are relying heavily on automation in order to make them effective and efficient for organizations to use.
- Compliance and security - configuration management can also assist an organization in meeting compliance and security standards across the entire organization.
- Documentation - there needs to be a fully documented process for both the configuration and change management procedures. Also, the configuration management system is often helpful or even fully responsible for helping to create accurate network documentation.
- Inventory management - many configuration management systems also double as inventory management systems. These are able to provide management with accurate records involving the hardware and software inventories that are available in the organization.
- Centralized management - most configuration management systems feature a centralized management approach. This permits engineers to visit a single source for any configuration or change management needs.
- Real-time monitoring - many configuration management systems have the ability to monitor network configurations in real time.

Additional Resources:

Configuration and Change Management
https://www.cisa.gov/sites/default/files/c3vp/crr_resources_guides/CRR_Resource_Guide-CCM.pdf